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ENTERPRISE SOLUTION PROPOSAL

# A multilingual claims assistant for India's health insurers

Resolving cashless, reimbursement and denial conversations in the caller's own language — voice, documents and grievance handling on one sovereign Indic AI stack.

Voice · Saaras + Bulbul

Reasoning · Sarvam-105B

Documents · Sarvam Vision

**₹1.1L cr**
Health insurance premium, FY25

**3.26 cr**
Health claims processed a year

**22 languages · code-mixed**
Where an English-first stack underperforms — and Sarvam wins

PICKING THE RIGHT PROBLEM

# We looked at five ideas. Helping people with health insurance claims was the clear winner.

| Use case | What the old way costs | Needs Sarvam | Space is open | Someone will pay | Verdict |
|---|---|---|---|---|---|
| **Helping people with health insurance claims** | ₹40–80 a call, plus repeat document work | High<br>Claims talk + hospital papers in many languages | High<br>No Sarvam customer here yet | High<br>Insurers hold the budget; savings are clear | **Build this** |
| **Turning doctor calls into medical records** | ~5 min of a doctor's time per visit | High<br>Mixed-language clinical speech | Low<br>HealthPlix already does this on Sarvam | High<br>Clear saving on doctor time | Strong, but taken |
| **Helping customers with loans and credit cards** | ₹40–80 a call | Medium<br>Often works in English or Hindi | Low<br>Sarvam already does BFSI here | High<br>Banks have budgets | Close, but crowded |
| **Answering shopper queries after they order** | ₹30–60 a call | Medium<br>English is often enough | Low<br>Many rivals already here | Medium<br>Thin retail margins | Pass |
| **Answering passport helpline calls** | ₹40–80 a call | High<br>Many languages needed | High<br>No AI player here yet | Medium<br>Slow government buying | Pass |

Health insurance claims is the only one where a **private company holds the budget**, the work **truly needs Indian languages and document reading**, and **Sarvam has no customer yet**.

WHERE CLAIMS BREAK DOWN

# A health claim isn't one conversation — it's three. People get stuck at each one.

Every year India handles 3.26 crore health claims. Most of the calls that follow fall into three kinds — and each one happens in a language an English-first system handles badly.

### 1 Cashless

**Questions before treatment**

Someone is being admitted and calls to ask if the hospital is covered and how to get cashless approval — anxious, and in their own language.

### 2 Reimbursement

**Filing after treatment**

They paid themselves and call to claim the money back. The claim stalls because a document is missing or unclear — often the discharge summary.

### 3 Denial

**A rejected or short-paid claim**

The claim was rejected or paid less than expected, and they call upset — because no one explained why in a way they understand.

THE CURRENT APPROACH

# Insurers handle these calls today — but no tool was built for how India speaks.

There are three ways these calls get answered right now. Each one lets the claimant down, so people fall back on costly human agents anyway.

### Human agents

- Costly — roughly ₹40–80 for every call
- Hard to keep available 24/7 in every language
- Difficult to staff for huge, unpredictable volumes

### IVR menus

- "Press 1, press 2" is frustrating — worse when you're anxious
- Can't read or check documents
- Can't reason or answer a real question

### Chatbots today

- Mostly English-first — which leaves out most of the country
- Can't understand or speak the way most Indians actually talk
- Can't handle documents

**The tools were built English-first and text-first — for a customer who is neither.**

BUILT FOR THIS PROBLEM

# This isn't a general AI problem — it's an India problem. Sarvam is built for it.

### Understands how India speaks

Listens and replies by voice in 22 languages, including mixed speech like Hindi-English.

Saaras · Bulbul

### Reasons through the claim

Checks the policy and explains cashless, status or a denial in plain words.

Sarvam-105B

### Reads Indian documents

Reads bills, prescriptions and discharge summaries — even handwritten, in regional scripts.

Sarvam Vision

### Keeps data safe in India

Runs inside the insurer's own environment, so sensitive health data never leaves their control.

Sarvam Arya

**A generic English-first stack does none of these four well enough for India.**

WHAT WE BUILT

# A voice assistant that handles the three claim calls — in the caller's own language.

A claimant reaches it from the **insurer's app** **(or a toll-free number in phase 2)**. It **listens, reasons, reads documents, and knows when to hand to a human** — all in the caller's own language.

### 1 Cashless

Explains coverage and pre-authorization, and checks status — in their language.

### 2 Reimbursement

Tells them what's needed, reads the uploaded documents, and flags what's missing on the call.

### 3 Denial

Explains the real reason in plain words, and hands to a human with the full context.

RUNTIME ARCHITECTURE

# One call, flowing through the system — the full Sarvam stack, working together.

CALLER ENTERS VIA  Insurer app · voice call  Toll-free · phase 2

**Speech in**
transcribes, auto-detects language
Saaras

**Orchestrator + reasoning** — finds intent, calls tools, composes the reply
Sarvam-105B · tool-calling

**FAST PATH · MOST CALLS**
Known answer → retrieve the exact fact → translate into the caller's language. No heavy reasoning.

**REASONING PATH · OPEN QUESTIONS**
Unusual question → Sarvam-105B reasons over the grounded facts and composes a reply.

TOOLS IT CAN CALL
get_claim_status · get_cashless_status · get_policy_clause · check_document_completeness · create_grievance_case

**Speech out**
speaks the reply in their language
Bulbul

**Insurer claim data + policy rules**
facts retrieved, never invented

**Reads uploaded documents**
OCR on bills, discharge, PAN
Sarvam Vision

**Human handoff**
denial or grievance → agent, with case summary

Everything runs **inside the insurer's own environment** — private, on-premise or fully sealed — with **every turn logged and audited**. Health data never leaves. Sarvam Arya

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THE ENGINEERING DECISIONS

# A demo is easy. Making it fast, safe and honest for real claims is the engineering.

1

PERFORMANCE

### We beat the latency tax

The model always reasons before answering — slow, and can't be turned off. So it **only picks a tool**; a "let me check that" clip plays **in parallel** with the lookup, pre-fetched a turn early. Zero dead air.

2

SAFETY

### The model never states a fact

It only **picks which tool to call** — it never sees or writes the final wording of a claim status or denial reason. Facts come from a lookup, worded by a fixed template. Each answer is tagged **"sourced from claim CL-1003."**

3

SCOPE

### We kept it lean on purpose

At one insurer's scale, **no vector database and no agent framework** — that would be over-engineering. We even replaced the vendor's turn-detection, which cut callers off mid-sentence, with our own.

4

DEPLOYMENT

### We designed the integration seams

Clear lines between **what's Sarvam, what's the insurer's** (claims system, CRM) and what's the integrator's — so this can be deployed into a real insurer, not just demoed.

WE TRIED TO BREAK IT

# The interesting failures aren't crashes — they're wrong answers that look right.

Adversarial testing found three subtle failures a casual demo would never surface. Each was fixed.

| WHAT BROKE UNDER TESTING | HOW IT WAS FIXED |
|---|---|
| **The right record, but the wrong one**<br>A question about "pet surgery" matched the human "cosmetic surgery" exclusion — a real record, retrieved wrongly. The hardest kind of error to spot. | **Raised the match threshold**<br>A single shared word no longer counts — a match now needs **two or more** overlapping terms. |
| **It answered things it shouldn't**<br>Asked for the weather or a joke, the model would sometimes just reply — breaking character. Exactly what a reviewer probes first. | **Added an out-of-scope catch-all**<br>Anything unrelated to the caller's policy, claim or documents now hands to a human, by rule. |
| **Escalation didn't always fire**<br>The trigger that hands a caller to a human was firing unreliably on some phrasings. | **Made the trigger deterministic**<br>Denials, disputes and human requests now escalate by fixed rule, not model judgment. |

The takeaway isn't the three bugs — it's that **a probabilistic system needs a standing adversarial test pass**, not a one-time check. That's how you run it in production.

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WALKTHROUGH — CASHLESS

# "Will my father's hospital be covered?" spoken in Hindi

THE CONVERSATION

CALLER HINDI

मेरे पिताजी को Aster अस्पताल में भर्ती कर रहे हैं। क्या ये cashless होगा?

*My father is being admitted at Aster hospital. Will this be cashless?*

ASSISTANT HINDI

जी हाँ, Aster एक network अस्पताल है। Cashless में बीमा कंपनी सीधे अस्पताल को भुगतान करती है। अस्पताल का desk pre-authorization भेजेगा।

*Yes, Aster is a network hospital. In cashless, the insurer pays the hospital directly. The hospital desk will send a pre-authorization.*

CALLER HINDI + ENGLISH

Approval में कितना time लगेगा?

*How long will the approval take?*

ASSISTANT HINDI

Planned admission में कुछ ही घंटे। आपकी request अभी submit हो चुकी है और pending है।

*A few hours for a planned admission. Your request is now submitted and pending.*

WHAT THE SYSTEM DOES

**Saaras hears the call**
Detects Hindi automatically and transcribes the code-mixed speech.

**Sarvam-105B finds the intent**
Cashless guidance → calls `get_cashless_status` and `get_policy_clause`.

**Facts come from the insurer's data**
Network hospital: yes · pre-auth: pending. Retrieved, never invented.

**Bulbul speaks the reply**
Answered in Hindi — the language the caller used.

WALKTHROUGH — REIMBURSEMENT

# "I paid myself — how do I claim it back?" spoken in Malayalam

## THE CONVERSATION

**CALLER** MALAYALAM

ഞാൻ ചികിത്സയ്ക്ക് പണം കൊടുത്തു. അത് തിരികെ കിട്ടാൻ എന്ത് ചെയ്യണം?

*I paid for the treatment myself. What do I do to get it back?*

**ASSISTANT** MALAYALAM

bill, discharge summary, prescription, PAN എന്നിവ വേണം. ഞാൻ ഒരു link അയക്കാം — അത് upload ചെയ്യൂ.

*You'll need the bill, discharge summary, prescription and PAN. I'll send a link — please upload them.*

**ASSISTANT** MALAYALAM

bill-ഉം prescription-ഉം കിട്ടി. പക്ഷേ discharge summary-യും PAN-ഉം കിട്ടിയിട്ടില്ല.

*I have the bill and prescription. But the discharge summary and PAN are still missing.*

## THE DOCUMENT SCAN

A secure link opens an upload screen — the caller scans two documents

**PRS HOSPITAL** OCR
Thiruvananthapuram, Kerala

**FINAL BILL / INVOICE**

| | |
|---|---|
| Patient Name | **Anil Kumar** |
| Bill No. | **PRS/2026/04821** |
| Admission | **14 Jun 2026** |
| Discharge | **17 Jun 2026** |
| Diagnosis | **Acute appendicitis** |

**CHARGES**

| | |
|---|---|
| Room charges (3 days) | Rs 15,000 |
| Surgery & OT | Rs 28,000 |
| Investigations | Rs 9,500 |
| Pharmacy | Rs 6,200 |
| Consultation | Rs 3,300 |
| **TOTAL** | **Rs 62,000** |

Paid by patient | Cashless: No | Reimbursement

✓ Hospital bill — read

**Dr. Meera Nair, MS** OCR
General Surgery • PRS Hospital

**Rx**

Patient: Anil Kumar Age: 42

Date: 17 Jun 2026

- **Tab. Augmentin 625mg**
  1-0-1 x 5 days
- **Tab. Pan-D 40mg**
  1-0-0 x 5 days
- **Tab. Dolo 650mg**
  SOS, if fever
- **Syp. Cremaffin**
  10ml at night

Review after 1 week.

Dr. Meera Nair
Signature

✓ Prescription — read

## WHAT THE SYSTEM DOES

**COMPLETENESS CHECK**

- ✓ Hospital bill
- ✓ Prescription
- ✗ Discharge summary — missing
- ✗ PAN for payout — missing

### UNDER THE HOOD

**Sarvam Vision reads the images**
Extracts fields from the bill and prescription.

**Checks against the checklist**
`check_document_completeness` flags what's missing.

**Bulbul says what's missing**
Answered in Malayalam, on the same call.

WALKTHROUGH — DENIAL

# "Why was only half my claim approved?" spoken in Hindi

## THE CONVERSATION

**CALLER** HINDI

मैंने ₹1,20,000 का claim किया था, लेकिन सिर्फ ₹75,000 approve हुआ। ऐसा क्यों?

*I claimed ₹1,20,000, but only ₹75,000 was approved. Why is that?*

**ASSISTANT** HINDI

आपकी policy में room rent की सीमा ₹5,000 प्रतिदिन है। कमरा उससे महंगा था, इसलिए उससे जुड़े charges उसी अनुपात में कम किए गए।

*Your policy has a room rent limit of ₹5,000 per day. The room cost more, so the related charges were reduced in the same proportion.*

✓ from an approved template — not invented

**CALLER** HINDI

मुझे ये सही नहीं लग रहा। मुझे किसी से बात करनी है।

*This doesn't seem right to me. I want to speak to someone.*

**ASSISTANT** HINDI

मैं आपकी शिकायत दर्ज कर रहा हूँ और एक agent से जोड़ रहा हूँ — उन्हें पूरी जानकारी मिल जाएगी।

*I'm logging your grievance and connecting you to an agent — they'll have the full context.*

## THE HUMAN HANDOFF

### Case passed to an agent — ESCALATED

CASE SUMMARY GENERATED AUTOMATICALLY

| | |
|---|---|
| Caller | Priya Menon · policy SMP-HL-500126 |
| Language | Hindi |
| Claim | ₹1,20,000 claimed · ₹75,000 approved |
| Explained | Room-rent cap deduction |
| Why escalated | Caller disputes the outcome |

### UNDER THE HOOD

**Denial reason comes from a fixed template**
`get_denial_reason` → room-rent cap. The model only phrases it.

**Dispute triggers escalation**
Denial + caller pushback → hand to a human, by design.

**Case summary handed over**
`create_grievance_case` — the agent starts with full context.

WHERE THE VALUE COMES FROM

# The win isn't a cheaper call. It's availability, and the rework it removes.

On a multi-minute claims call the per-call cost is roughly comparable to a human. The real value is elsewhere — and it's larger.

THE BIGGEST SAVING

**1 in 3**

**document resubmission cycles removed — the missing paper is caught on the very first call.**

Each avoided cycle saves several follow-up calls and weeks of delay — a far bigger cost than any single call.

### Available 24/7, in every language

No human can answer in Malayalam at 2am. The assistant can — **no staffing for spikes, no language gaps**.

### Deflects the simple, short calls

Pure status and cashless-info calls are quick and cheap for AI — **freeing agents for the hard cases**.

### Fewer grievances, better retention

Clear answers in the caller's own language reduce disputes — **and the customers who leave over them**.

*Honest note: on a 3–4 minute claims call, AI (~₹15–30) and a tier-2 human agent (~₹40–50) are broadly comparable per call. The saving comes from availability, deflecting short calls, and removing rework — not from a large per-call price gap. Figures are illustrative ranges for sizing, not audited results.*

HONEST CONSTRAINTS

# We know exactly where the edges are — and the path past each one.

| WHERE THE POC STOPS TODAY | | HOW IT GETS TO PRODUCTION |
|---|---|---|
| 1 | **It isn't instant**<br>The reasoning stack adds a few seconds per turn. | **Stream replies + real-time voice**<br>The fast/slow path routing already helps; Pipecat streaming closes the gap where speed matters. |
| 2 | **It runs on mocked claim data**<br>The PoC proves the conversation, not live lookups. | **Connect the real claims system + CRM**<br>Through the integration seams already designed into the architecture. |
| 3 | **The knowledge base is small**<br>One insurer's rules fit in context — no retrieval layer yet. | **Add retrieval (RAG) at scale**<br>A vector layer when it spans many products and insurers. |
| 4 | **Denial wording is translated from English**<br>Fine for a demo, but denial text is legally sensitive. | **Human-reviewed vernacular templates**<br>Approved wording in each language for regulated messages. |
| 5 | **OCR reads documents, but doesn't judge them**<br>It extracts text; it doesn't yet catch every messy or unclear scan. | **Human check on unclear documents**<br>Low-confidence scans routed to a person, not auto-accepted. |

*Stated plainly on purpose — a PoC that names its limits is one you can trust in production.*

FROM WEDGE TO SCALE

# Start with three calls at one insurer. Grow into the whole claims-servicing layer.

THE EXPANSION PATH

START HERE

### This PoC

Three claim calls, one insurer, on Sarvam's stack — small, provable, safe.

THEN

### More calls, more insurers

Add claim types and roll the same design out to more insurers and TPAs.

SCALE TO

### The full servicing layer

The multilingual voice + document layer across the whole claims journey.

WHY IT'S THE RIGHT BEACHHEAD FOR SARVAM

**Open white space**

No Sarvam customer in health-claims servicing yet.

**Uniquely Sarvam**

Needs exactly its Indic voice, documents and sovereignty.

**A real buyer**

Private insurers with budgets and a felt cost.

THE NEXT STEP

### A paid pilot with one insurer

One product, two languages, the three calls — with success metrics agreed up front and a clear gate to rollout.

DEPLOYMENT & INTEGRATION

# It runs inside the insurer's walls — and plugs into the systems they already have

INSIDE THE INSURER'S SECURE ENVIRONMENT

THE ASSISTANT

## Sarvam voice + document + reasoning core

Saaras · Bulbul · Sarvam-105B · Vision · Arya

PLUGS INTO THE INSURER'S OWN SYSTEMS

- **Core claims system** — live claim status and history — Insurer
- **Policy & product master** — coverage, limits, waiting periods — Insurer
- **CRM / ticketing** — the human-agent handoff — Insurer
- **Telephony** · phase 2 — toll-free reach beyond the app — Integrator

**Private cloud (VPC)**
in the insurer's own tenancy

**On-premise**
on the insurer's own servers

**Fully sealed / air-gapped**
no external network at all

HOW IT LANDS AND SCALES

# Start with a paid pilot, prove it against agreed metrics, then scale on evidence.

**PHASE 1**

### Paid pilot

One insurer, one product, two languages, the three claim calls. Tight scope, real users.

**PHASE 2 · THE GATE**

### Prove the metrics

Measured against targets agreed up front. A clear go / no-go before any rollout.

**PHASE 3**

### Production rollout

Connect the live claims system, add telephony, widen to all callers of that product.

**PHASE 4**

### Scale out

More claim types, more products, more insurers — the full servicing layer.

THE GATE CRITERIA — already measured by the PoC's own metrics dashboard, so the pilot reports them from day one

**Containment rate**

share of calls resolved without a human

**Docs right first time**

uploads read and classified on first try

**Language-wise quality**

containment holds across each language

**Caller satisfaction**

thumbs-up rate on resolved calls

The pilot isn't a leap of faith — **the same metrics that gate production are already instrumented in the PoC.**

COST DISCIPLINE

# Every turn was designed to spend as little as it can get away with.

On a reasoning-heavy stack, cost is a design problem. Six choices keep the per-call spend down.

### The model only picks a tool

It makes a small **50–150 token decision** per turn — not composing long answers. The expensive reasoning is paid once, and kept tiny.

### Translate, don't reason, to localize

Answers are worded once in English, then **translated per turn** — a cheap translation call, not a second expensive reasoning call.

### The fast path skips reasoning

Known, routine answers **never touch the reasoning model** at all — retrieve, translate, speak.

### Facts come from free lookups

Claim status, coverage and denial reasons are **Python lookups**, not paid model calls. The facts cost nothing.

### No always-on infrastructure

One service, **no database, no vector store**. No embedding costs, no query engine humming when no one's calling.

### The filler clip is made once

The "let me check that" audio is **synthesized a single time and cached** — not regenerated on every call.

QUALITY DISCIPLINE

# Every choice here exists to make the answer one you can trust.

Trust isn't a prompt — it's built from specific, testable decisions. Six of them.

### The model never states a fact

Facts come from a lookup and a fixed template — so a coverage rule or denial reason **can't be hallucinated**.

### We tried to break it

Adversarial testing caught subtle failures — a **wrong-record match**, out-of-scope replies — and each was fixed.

### Document reading is bounded

The OCR classifier is **forced to pick from a fixed list** of expected documents — never a free-text guess.

### Spoken IDs still resolve

"SMP HL 500123" and "smphl500123" both find the right claim — IDs are **normalized**, so speech quirks don't break lookups.

### We replaced the vendor's turn-detection

Sarvam's own end-of-speech signal cut callers off mid-sentence — so we **built our own**, tuned to real pauses.

### Answers show their source

Each reply is tagged **"sourced from claim CL-1003"** — so a reviewer can verify it was retrieved, not invented.